11/6/2026

Last time:

$$\pi(N) > \frac{1}{4} \cdot \frac{N}{\ln N}, \quad N \geq 2.$$

Upper bound.

Look at

$$\binom{2n+1}{n} = \frac{(2n+1)!}{n!\,(n+1)!}$$

$$= \frac{(n+2)\cdots(2n+1)}{n!} \in \mathbb{Z}.$$

1. $\prod_{n+2 \leq p \leq 2n+1} p \,\Big|\, \binom{2n+1}{n} \Rightarrow \prod_{n+2 \leq p \leq 2n+1} p \leq \binom{2n+1}{n}.$

$$\Rightarrow \sum_{n+2 \leq p \leq 2n+1} \ln p \leq \ln \binom{2n+1}{n}.$$

2. $\binom{2n+1}{n} \leq 2^{2n} = 4^n$.

3. Define

$$\theta(x) = \sum_{p \leq x} \ln p .$$

$$\sum_{n+2 \leq p \leq 2n+1} \ln p \leq \ln \binom{2n+1}{n} \leq 2n \ln 2$$

$$\theta(2n+1) - \theta(n+1)$$

So

$$\theta(2n+1) - \theta(n+1) \leq 2n \ln 2 , \quad n \geq 1 .$$

We prove by induction that

$$\theta(n) < 2n \ln 2, \quad n \geq 1.$$

1. $n = 1$: $\theta(1) = 0 < 2 \cdot 1 \cdot \ln 2$. ✓

2. $n = 2$: $\theta(2) = \ln 2 < 2 \cdot 2 \ln 2$. ✓

3. Assume $\theta(n) < 2n \ln 2$ for all $n \leq n_0 - 1$, where $n_0 \geq 3$.

① n_0 is even :

$$\theta(n_0) = \theta(n_0 - 1) < 2(n_0 - 1) \ln 2 < 2 n_0 \ln 2. \quad ✓$$

$n_0 \geq 4$ even

$\Rightarrow n_0$ is composite

② n_0 is odd: Write $n_0 = 2m+1$, $m \geq 1$.

$$Q(n_0) = Q(2m+1) = \underbrace{Q(2m+1) - Q(m+1)}_{\leq 2m \ln 2} + \underbrace{Q(m+1)}_{< 2(m+1)\ln 2}$$

because $m+1 < n_0 = 2m+1$.

$$< 2(2m+1)\ln 2$$

$$= 2n_0 \ln 2.$$

By induction, we are done.

Now,

$$2N\ln 2 > \theta(N) = \sum_{p \le N} \ln p \geq \sum_{N^c < p \le N} \ln p \quad , \quad 0 < c < 1.$$

$$\geq \sum_{N^c < p \le N} \ln N^c .$$

$$= c \ln N \sum_{N^c < p \le N} 1$$

$$\pi(N) - \pi(N^c)$$

Solving for $\pi(N)$:

$$\pi(N) < \pi(N^c) + \frac{2\ln 2}{c} \cdot \frac{N}{\ln N} .$$

$$\leqslant N^c + \frac{2\ln 2}{c} \cdot \frac{N}{\ln N} .$$

$$\pi(N) < N^c + \frac{2\ln 2}{c} \cdot \frac{N}{\ln N}$$

$$= \left(\frac{\ln N}{N^{1-c}} + \frac{2\ln 2}{c} \right) \frac{N}{\ln N}$$

Take $c = \frac{2}{3}$. Then

$$\pi(N) < \left(\frac{\ln N}{N^{\frac{1}{3}}} + \frac{2\ln 2}{2/3} \right) \frac{N}{\ln N}$$

$$= \left(\underbrace{\frac{\ln N}{N^{\frac{1}{3}}} + 3\ln 2}_{\leq 3.18\ldots \text{ for all } N \geq 2} \right) \frac{N}{\ln N}$$

$$< \underbrace{3\frac{1}{4}}_{\frac{13}{4}} \cdot \frac{N}{\ln N}$$

Hence,

$$\boxed{\frac{1}{4} \cdot \frac{N}{\ln N} < \pi(N) < \frac{13}{4} \cdot \frac{N}{\ln N}, \quad N \geq 2.}$$

Ex. Find the value $c \in (0,1)$ that minimizes the global max of

$$\frac{\ln N}{N^{1-c}} + \frac{2\ln 2}{c} \quad \text{on} \quad [2,\infty).$$

By working harder, we can prove

Thm (Bertrand's postulate) We have

$$\pi(2N) - \pi(N) \geq 1, \quad N \geq 1$$

In other words, there is always a prime $p \in (N, 2N]$.

Arithmetic functions

An arithmetic function is a function f with domain being the set of positive integers, denoted by $\mathbb{N}$, and range being a subset of $\mathbb{R}$ or $\mathbb{C}$:

$$f : \mathbb{N} \longrightarrow \mathbb{R} \text{ or } \mathbb{C}.$$

In particular, $\{f(n)\}_{n=1}^{\infty}$ is a sequence.

Eg. 1. $f(n) = 1$, $n \in \mathbb{N}$

2. $f(n) =$ number of positive divisors of n.

$f(6) = 4$, divisors of 6 are: 1, 2, 3, 6.

3. $f(n)$ = sum of positive divisors of n.

$f(6) = 1+2+3+6 = 12 = 2 \cdot 6$.

4. $\pi(n)$ = number of primes $\leq n$.

5. $f(n) = \begin{cases} 1, & \text{if } n \text{ is prime,} \\ 0, & \text{otherwise.} \end{cases}$

So $\pi(N) = \sum_{n=1}^{N} f(n)$.